



Routledge Advances in Urban History

POLITICS OF URBAN KNOWLEDGE

**HISTORICAL PERSPECTIVES ON THE SHAPING
AND GOVERNING OF CITIES**

Edited by
Bert De Munck and Jens Lachmund



Politics of Urban Knowledge

This book uses ‘politics of urban knowledge’ as a lens to understand how professionals, administrations, scholars and social movements have surveyed, evaluated and theorised the city, identified problems and shaped and legitimised practical interventions in planning and administration.

Urbanisation has been accompanied, and partly shaped by, the formation of the city as a distinct domain of knowledge. This volume uses ‘politics of urban knowledge’ as a new perspective on urban history and urban planning history. Through case studies of mainly 19th and 20th century examples, the book demonstrates that urban knowledge is not simply a neutral means to represent cities as pre-existing entities, but rather the outcome of historically contingent processes and practices of urban actors addressing urban issues and the power relations in which they are embedded. It shows how urban knowledge-making has reshaped the categories, rationales and techniques through which urban spaces were produced, governed and contested, and how the knowledge concerned became performative of newly emerging urban orders.

The volume will be of interest to scholars and students in the field of urban history and urban studies, as well as the history of technology, science and knowledge and of science studies.

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Contents

<i>List of figures</i>	vii
<i>Acknowledgements</i>	x
<i>List of contributors</i>	xi
1 Introduction: Urban knowledge and the politics of governing cities	1
BERT DE MUNCK AND JENS LACHMUND	
PART I	
Ways of Urban Knowing	27
2 The emergence of cartographic reasoning in a long-term perspective: Urban knowledge, craft corporations and body politics	29
JASNA SERŠIĆ AND BERT DE MUNCK	
3 Epistemological fields of urban intervention: Urban reform, surveys and historic centres in the late nineteenth and early twentieth century	56
PIETER UYTENHOVE AND WOUTER VAN ACKER	
4 From the ‘scientised’ to the ‘sociocratic’ city: The politics of knowledge and norm change in post-war urban planning in the Netherlands	81
TIM VERLAAN AND STEFAN COUPERUS	

PART II

Trajectories of Urban Knowing 109

- 5 Urban populations and urban problems
in Quetelet's population statistics of the
mid-nineteenth century 111
KAAT LOUCKX
- 6 Knowledge appropriation in Belo Horizonte:
Intertwining the urban, the suburban and the rural 133
PATRICIA CAPANEMA ALVARES FERNANDES AND
VIVIANA D'AURIA
- 7 Decoding zoning: Categorisation and commonality
in land-use planning knowledge 157
JULIO PAULO AND MARKO MARSKAMP
- 8 Between straight lines and winding alleys:
Streets as boundary objects in the transnational
modernisation of urban planning in Iran 179
POUYA SEPEHR AND ERIK AARDEN
- 9 Counting and caring for urban trees: Street tree
surveys and citizen protest in twentieth-century
West Berlin 201
SONJA DÜMPELMANN
- 10 Knowledge-making and the quest for a sustainable
city: Promoting community food-growing in
London 229
JENS LACHMUND
- 11 Smart cities, knowledge generation and the
enduring pursuit of urban innovation 253
ANDREW KARVONEN
- Index* 270

Figures

3.1	a and b: Two publications of the Musée Social: ‘Les classes moyennes et le logement’, a lecture of Georges Risler, published in <i>Le Musée Social: Mémoires et Documents</i> , March-April 1921, and the nr 8, August 1899, issue of <i>Musée Social</i> , its monthly journal. (Gallica, Bibliothèque nationale de France.)	62
3.2	Victor Bourgeois, plans and perspective of the Urbaneum as conceived in collaboration with Paul Otlet, 1931. Archives d’Architecture Moderne, Ixelles	69
3.3	R. De Cramer, Gand 1913. Vieille Flandre at the Universal Exhibition, Hélotypie De Graeve, Ghent. https://lib.ugent.be/catalog/rug01:001011170	72
4.1	Planners standing over the Hoog Catharijne scheme: a clear example of how the ‘scientised city’ regime promoted a technocratic approach to urban planning	88
4.2	An action group meeting attended by two young architects: a clear example of how the ‘sociocratic knowledge regime’ came to protest the ideas underpinning the scientised city	94
4.3	A schematised depiction of Hoog Catharijne’s traffic routes	96
4.4	Visual montage of how office developments threatened the urban fabric, produced by a community action group	99
5.1	Map illustrating poverty (‘indigence’) in the city of Brussels (Quetelet, 1844, appendix, s.p.)	118
6.1	Sample from the Study Commission final report, showing the volume of water available in each of Belo Horizonte’s creeks, before intervention	137
6.2	Geodesic, Topographic and Cadastral Plan for the studied zone. Scale 1:10.000. Comissão Constructora da Nova Capital. March, 1895. First Engineer Hermínio Alves	139

6.3	Planta Geral da Cidade de Minas by Aarão Reis, of 1895	142
6.4	Urban, suburban and rural zones, superimposed by the agricultural colonies implemented thereafter	143
6.5	Superimposition of Aarão Reis' plan of 1895 with the resultant urban form in 1928, the demolished village (Arraial), the plot reservation and the agricultural colonies implemented	146
6.6	Original plot subdivision (top) and Cadastral map of 1942 (bottom) in Section VIII (Vila Operária, left) and Sections V and VI (Bairro dos Funcionários, right)	148
6.7	Occupation according to 1928s Cadastral map. Between parallel lines, the strip for 30,000 inhabitants. In lighter red the urban zone and darker the suburban. The northwest section in the urban zone corresponds to the Vila Operária area. In green the agricultural colonies implemented from 1895	150
7.1	Zoning map in 1928 Vancouver plan by Bartholomew and associates	164
8.1	Sanandaj master street plan, as published in Alten (1958)	189
8.2	Esfahan Master Land Use Plan, with differently accentuated functional zones. Published in Kocks et al. (1961)	190
8.3	The changing street plan of Kashan. Published in Sadri (2014: 29)	193
9.1	Images provokingly juxtaposing a healthy mature tree, and a tree in a planting container used to protect newly planted trees against de-icing salt. From a brochure titled 'Aktion Tausalzstopp. Pro und kontra Streusalz' issued by Arbeitsgruppe Wasser im Bundesverband der Bürgerinitiativen Umweltschutz e.V., December 1979. Private collection Jürgen Hübner-Kosney	207
9.2	Front page of a brochure issued by the citizen initiative Aktion Tausalzstopp against the use of de-icing salt, 1980. Private collection Jürgen Hübner-Kosney	209
9.3	Top: Election campaign poster of West Berlin's Christian Democratic Party (CDU), 1985. Landesarchiv Berlin, F Rep. 260-03 Nr. B_1048. Bottom: 'Where are the new trees?' <i>Berliner Luft-Zeitung</i> , no. 13 (1989)	214
9.4	Top and middle: Two versions of protective shields against dog urine in Berlin. Bottom: Huto in Westerland, Sylt. Top photo: Schaefer, middle and bottom photo: Hartmut Balder. (Balder, 1990: 738)	216

9.5	Election campaign poster of West Berlin's Alternative Liste für Demokratie und Umweltschutz (AL), 21 May 1985, West Berlin. Photo: Jürgen Henschel. FHXB Friedrichshain-Kreuzberg Museum, K03_0469_08	217
10.1	Food growing in a London community garden: Streatham Common 2015. Photo by the author	232
11.1	Contemporary street art in Manchester celebrating the iconic 'worker bee' from the city's Industrial Era	256
11.2	Projects in MCC's Manchester Smarter City Programme	259
11.3	An image of the Rochdale Canal with Royal Mill and Sedgwick Mill in the background	262

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1 Introduction: Urban knowledge and the politics of governing cities

Bert De Munck and Jens Lachmund

There has been much debate recently, in urban studies and popular literature, about the supposedly increasing role of knowledge in the contemporary ways of governing and planning cities. This includes work on a move towards ‘knowledge-based’ models of governance (Anderson and Atkinson, 2013), emerging ‘knowledge-economies’ (Florida, 2002, 2004), and the increasing popularity of novel formats of public experimentation (Karvonen and van Heur, 2013; Evans et al., 2016). Likewise, mounting work on ‘smart’ cities has documented the actual, or potential, penetration of urban life by complex digital dataspace (Coletta et al., 2018; Karvonen et al., 2018). Such diagnoses have been accompanied both by optimistic visions of increasing civic participation, socio-economic ‘creativity’ and sustainability, as well as by warnings about the rise of technocratic forms of power (Raco and Savini, 2019).

From a historical perspective, the knowledge-based character of contemporary urban governance and politics is not new. At least since early modernity, the worldwide process of urbanisation has been accompanied and partly shaped by the formation of the city as a distinct domain of knowledge. Cities provided environments whose dynamics and complexity were unprecedented, and knowledge-making was key when planners, experts and policy makers, as well as urban publics tried to make sense of its variegated phenomena and problems. Such ‘urban knowledge’, as we call it in this book, took shape in different media, such as medical reports, topographical surveys, municipal statistics, social policy surveys and the research projects of urban sociologists and geographers, to name just a few. It materialised in a host of administrative, governmental and civic practices, as well as in the expertise of new professions. It provided the tools for the diagnosis and explanation of the supposed pathologies of the modern city, and it guided and legitimised attempts at political control and reform.

This book is about the constitution and authority of such diverse forms of urban knowledge, and the political roles that they have played throughout the history of the city. We seek to understand how urban knowledge has been shaped, how it has been validated and contested and

how it has circulated among different cities. At the same time, the book traces how such knowledge has been entangled with the problematisation of urban conditions and with the articulation of visions and strategies of urban reform. Conceptually, we thereby advance a perspective of knowledge as a situated process of meaning-making and practical action.

It is our first and most immediate goal in this collection to shed light on the role of knowledge-practice as a dimension of city-making. *As we claim, knowledge of the city does not simply register pre-given urban realities and problems but is recursively tied to the development of the city itself. Urban knowledge, in other words, is performative.*¹ Whether it is the built form of the city, its socio-economic institutions, or the technologies and machineries of its governing – all of these embody, to a varying extent, the intellectual and material products of particular knowledge processes. A second, and complementary goal of this project is to ground urban knowledge in the concrete historical dynamics and power configurations in which it evolved. As much as cities materialise urban knowledge, such knowledge bears the imprint of the institutions and contexts in which it is created and of the mentalities, interests and commitments of the involved researchers and mediators. Our project therefore sets out to relate the history of ideas on and knowledge of the city to the actual dynamics and trajectories of the historical cities in which the guiding ideas and visions of planning and governance have emerged. *This implies that the history of urban knowledge can provide a specific lens through which to look at the long-term historical development of cities – one which reveals the twists and turns and the continuities and discontinuities that are easily neglected by linear narratives and periodisations of urban history.* By tracing their urban genealogies, the book also attempts to historicise categories, narratives, models and actors that we tend to see as universal and which still survive in such disciplines as urban sociology, urban geography, architecture, urban planning, urban environmental studies, etc.

Historians have pleaded recently for a turn towards a ‘history of knowledge’, which seeks to ground historical phenomena in contingent orders of knowledge (Dupré and Somsen, 2019; Sarasin, 2011; Vogel, 2004).² The approach taken in this book can be understood as an attempt to advance a similar move within urban history. Conceptually and methodologically, it also builds on work in the field of science and technology studies, the history of science, and notably on earlier work on an ‘urban history of science’ (Dierig et al., 2003). The focus, however, is not broadly on scientific knowledge processes that are located, and somehow shaped in cities, *but on how cities themselves are configured as objects of knowledge and related politics.* This includes urban knowledge processes within the domain of academic science; as well as the practices and discourses through which planning professions, urban

bureaucracies, political agencies, publics and urban movements made these cities known and, on this basis, organised collective action and institutional networks.

Our book thereby complements the self-reflective debate among critical urban researchers on the ‘politics of urban knowledge’ as it has recently been conducted in the journal *City*. As one contributor put it:

urban knowledge (...) is part of the field of contestation over urban space itself. Even the most technocratic and disembodied representation seemingly emanating from the view from nowhere has at least a crypto-political significance in that it establishes which measures are visible, knowable and hence valuable and which are not.

(Madden, 2015: 300)³

While the debate in *City* revolved mainly around urban study researchers’ positionality and epistemological choices, **we treat such politics of knowledge here as a theme of historical analysis.**

At the same time, by treating cities as constituted through the knowledge processes by which they are made known, we hope to avoid a naive ‘methodological cityism’ (Angelo and Wachsmuth, 2014), which treats cities as predetermined essential entities but neglects the extent to which urbanisation unfolds as a planetary process. If we refer to our objects in this book as ‘cities’, we do this with respect to the knowledge processes through which historical actors identified these places as such, and, explicitly or implicitly, differentiated them from other spatial configurations – not as entities which are cities because of their specific intrinsic characters. In this respect, our view is not rooted in the political economy framework that has guided major work on ‘planetary urbanization’ (Brenner, 2013; Brenner and Schmid, 2015); **but it dovetails with recent attempts to view cities as relationally produced ‘actor-networks’, or ‘assemblages’ of people, material objects, institutions, rules, discourses, practices and – indeed – knowledge** (Fariás and Bender, 2010; McFarlane, 2011; De Munck, 2017; Van Damme and De Munck, 2017). Making cities known and knowable can be seen as one of the layers of relational practice through which such urban assemblages and agency come into being. As all the contributions show, this is never only a local process that is fully contained within single cities, but rather one which is articulated with multi-scalar processes of knowledge circulation and appropriation.

This collection brings together a broad range of historians with an eye for an integrated view, in which knowledge formation and epistemological transformations are confronted with the material, social and cultural orders of tangible cities. With the exception of the first chapter that traces the roots of these developments back to the Renaissance, we focus mainly on the nineteenth and twentieth centuries, a period of modernisation and industrialisation in Europe and North America in which

expert-based forms of urban planning and administration emerged, including the related counter-movements. Although we also consider the circulation to and interaction with urban histories in colonial and post-colonial contexts, we do not claim to provide a global history of urban knowledge in this volume. Expert-based urban planning has not only marked Western history, but has subsequently been adopted and applied in large parts of the world. It is a textbook example of how a typically Western relationship between knowledge and politics has become hegemonic and has ultimately marked the city and urban society worldwide. Some authors in this book build on this critique through insights, from post-colonial thinking, that expose the specific Western (or European) relationship between political and epistemological developments. In this respect, our book is also an attempt to ‘provincialise’ (Chakrabarty, 2000, esp. ch 2) the model of knowledge and the city as it has evolved in Europe.

In this introductory chapter, we will sketch the main ideas that guide our project of a knowledge history of the city and situate it in the existing landscape of research in urban history and urban studies. We will elaborate specifically on how we understand the *political* nature of urban knowledge. In the first section of this introduction, we will go deeper into the notion of politics of urban knowledge and try to indicate which historiographical and theoretical insights we build upon. We will also indicate which dimensions of the relationship between knowledge and politics we consider relevant and important. Next, we will elaborate on the contributions in the book and briefly discuss the insights that have emerged from them.

Urban knowledge and its politics

There is quite some scholarship in urban history and notably planning history, which has studied the role of experts and knowledge in the design and material shaping of cities. This includes countless planning histories of individual cities, which have detailed the interaction between planners, local policy makers and other stakeholders. There is also quite a substantial social and intellectual history of planning ideas and their promoters (e.g. Boyer, 1983; Hall, 2014; Ladd, 1990; Ward, 1992, 2002; Wakeman, 2016). Social historians have, in addition, often looked at the ways in which urban administrations have framed and governed urban problems. Especially relevant here are studies which have applied Foucauldian notions of power to analyse the rationalities of modern urban forms of governing (Rabinow, 1995; Joyce, 2003). In the meantime, work on colonial cities has also appeared, which studies the links between modern precepts of rational planning and the contestation of colonial power (Healey and Upton, 2010; Roy, 2009; Bigon, 2016; Barry and Thomsen-Fawcett, 2020). Although there is much overlap with our book, and we will later come back to

some of these publications in our review, it has not been the intention of such studies to make knowledge processes as such the focus of urban historical analysis. So, what is still missing is a more systematic attempt to zoom in at the internal complexity of the processes through which urban knowledge has been produced, valued and articulated by different urban actors, with different political agendas and in different urban contexts. By this we do not only refer to the formation and diffusion of general ideas and models of urban planning (Ward, 2002), but to whole sets of underlying practices and techniques that constitute the city as a field of knowledge and intervention. This also includes a sensitivity for the multiplicity of urban knowledge and its openness to negotiation and contestation.

An important source for a broader orientation is the work of historians of science and more broadly the field of science and technology studies. Since the 1980s, this work has problematised the linear history of the scientific progress. Rather than treating knowledge as a context-independent domain in which natural objects and facts are objectively represented; they have used detailed historical and ethnographic case studies to trace how knowledge claims were shaped, negotiated and stabilised (Knorr-Cetina, 1981; Latour, 1987; Latour and Woolgar, 1979; Collins, 1992). Such work has brought into focus how even the most technical forms of scientific knowledge bear the imprint of historically contingent epistemic normativities, power relations and local practicalities (e.g. Shapin and Schaffer, 1985; Shapin, 1994, 1996; Poovey, 1998). Notably, work on an ‘urban history of science’ (Dierig et al., 2003; Hochadel and Nieto-Galan, 2016, 2019) has provided an important stepping-stone for this book. In their *Osiris* volume on ‘Science and the City’, Dierig et al. (2003) have already identified four systematic ways in which scientific knowledge is co-produced with the social, material and political order of the city: expertise, representations, ‘places of knowledge’, and participation in urban everyday life.

Recent work on the ‘history of knowledge’ has moreover identified knowledge as a key dimension of any domain of social and cultural history, and thus extended such analytical interests beyond the domain of science. State and corporate bureaucracies are no longer only portrayed as formal organisations, but as technologies of knowledge and collective learning processes (Felten and Oertzen, 2020). Likewise, this book moves beyond an urban history of science to what can be characterised as a knowledge history of the city. Knowledge penetrates and constitutes cities in very different ways, up to and including in everyday practices of walking, in sharing a ‘sense of place’, or in practices of consumption. Thus, for example, in his *Reading Berlin*, Peter Fritzsche (1998) nicely sketched how, around 1900, newly arriving residents had to acquire complex practical forms of knowledge, and identified the local press as an important source of such knowledge.

The knowledge histories presented in this volume have a more limited focus: they set out to probe those forms and practices of urban knowledge that are mobilised in the processes of problematising, governing and contesting urban spaces, a field that we characterise broadly as ‘politics of urban knowledge’. This does not mean that the residential knowledge practices, that Fritzsche described, for example, did not also have a political dimension (e.g. were gendered or racialised). In this volume, however, we consider such knowledge practices only to the extent that they are mobilised in acts of resistance and collective mobilisation against political authorities, or vice versa, when professional and administrative forms of knowing seek to discipline them in line with governmental rationalities. Indeed, many of the contributions to this volume will deal systematically with intersections and tensions between different registers of urban knowledge.

In line with this, we will treat urban knowledge as profoundly processual. In contrast to work on stable urbanist ideas or Foucauldian abstractions of overarching diagrams of governmentality, we are interested in historically situated ‘knowledge processes’, i.e. sequences of practices, interactions, events and institutional transformations in which knowledge takes shape, is validated, contested, mobilised and gets materialised as tangible urban spaces and places. We use knowledge thereby to refer both to the use of representations (discourses, ideas, imaginaries, maps, etc.) of urban spaces and spatial phenomena, as well as the practical understandings and skills of the fields of social actions in which they are embedded. With our processual view of knowledge, we build on work informed by actor-network theory and other practice-oriented forms of social studies of science (Knorr-Cetina, 1981, 1999; Latour, 1987, 2007; Pickering, 1992, 1995). We understand these knowledge processes as situated, in the sense that they build on historically specific constellations of actors, institutions, discourses and materials that produce ‘truths’ about the city in locally contingent ways. As Gieryn (2007) has argued, specific cities, such as Chicago for early urban sociology, have even functioned as ‘truth spots’ that are intimately attached to certain intellectual projects and contribute to the credibility of knowledge claims. In the two parts of the book, we focus both on recurrent patterns (‘ways of knowing’) and specific local trajectories of such urban knowledge processes.

Our concern, however, is not only about the epistemic content of these processes, but about how they intersect with, or conflict with, processes and practices of urban politics and governance. In this sense, we build on work in science and technology studies which have extended the focus from the social production of knowledge to, as Jasanoff (2004) put it, the ‘co-production’ of science and social and political order.⁴ We want to understand how knowledge is produced within, and how it affords certain ways of doing urban politics, either in formal processes of

governing a city or in contentious public action that challenges or precedes such policies. Who is included and who is excluded from urban knowledge processes? What do they make visible and what do they make invisible? Who benefits from and who sponsors these processes? Who acquires the authority to speak in the name of the city or of urban phenomena, and what is the role of knowledge in this process? In all these respects, knowledge is involved as a contested field of political relations, whether implicitly or explicitly.

To further substantiate our argument, we distinguish four interrelated dimensions of such a co-productive interaction between knowledge, politics and the city. We identify how existing literature has already contributed to the understanding of these dimensions, and outline how we seek to further develop this into a more comprehensive history of urban knowledge.

Governing and administering the city

The first common theme running through these contributions is the role of knowledge as a technique for governing the city. Making and processing urban knowledge is a core feature of any attempt to govern a city. This is a lesson learned from James Scott (1998), who used the metaphor ‘seeing like a state’ to describe how urban governments have relied on standardised schemes and abstractions to make cities and their issues calculable and manageable in goal-directed policies. Others have drawn on Michel Foucault’s work, including his concept of governmentality, to analyse how knowledge-based power technologies such as statistics and maps have been involved, in various historical periods, in the operation of distinct regimes of shaping order (Osborne and Rose, 1999; Magnusson, 2011). Foucault’s concepts have been picked up in more substantial historical studies, such as Rabinow’s book *French Modern* (1995), which traces the role of urbanists and social reform experts in the nineteenth century and shows how technical competences were interrelated with the imposition of new norms in social life. Patrick Joyce has shown that the spatial and material shape of cities was also part of a broader governmentality in which experts and expertise mediated between ideals of governance, on the one hand, and the behaviour and subjectivity of urban dwellers, on the other. In his book *Rule of Freedom* (2003), Joyce revealed how ‘technologies’ as maps and plans as well as sewers and new streets were constitutive of a liberal type of governance in nineteenth-century English cities, as they were key in the production of self-governing subjects. In a similar fashion, Chris Otter argued in *The Victorian Eye* (2008) that the introduction of such technologies as gas and electricity and such material devices as plate glass and light switches helped shape a liberal form of governance in general and a liberal, self-governing subject in particular (see also Burchell et al., 1991; Bennett and Joyce, 2010).

State abstraction or ‘governmentalities’ imply important forms of urban knowledge, but this does not mean that they have power-effects in themselves. By definition they operate only in conjunction with other elements of a local historical situation and are always open to contestation and re-definition. Moreover, as Valverde (2011) argued, one should not underestimate the extent to which context-specific, pre-modern forms of political knowledge can still coexist with modern power-technologies in urban governance. Nor does work on state abstractions or governmentality have a great deal to say about the situated work of observing and surveying the city through which such political knowledge comes into being, and how their nexus with governmental practices is formed. There is nothing intrinsically instrumental in knowledge that makes it useful for governing (see also Jasanoff, 2004). Knowledge has to be tailored to the needs of governing, and that is often a painstaking process of mutual adaptation of knowledge practices and representations with the agendas and institutional-procedural environments of governmental organisations (Lachmund, 2013). Even standardisation is not an intrinsic feature of state knowledge that could make it more powerful than other forms of knowledge, but rather itself the result of social negotiation and tinkering (Bowker and Star, 2000; Epstein, 2006).

In line with our processual understanding of urban knowledge, we will zoom deeper in at the socially situated dynamics of making and negotiating knowledge that is implied in the practice of governing. In contrast to Foucauldian studies, which use historical evidence primarily to demonstrate overarching historical logics or ‘diagrams’ of power, the contributions to our book seek to ground such nexuses between knowledge and politics into locally specific stories, or long-term trajectories, that probe the dynamics through which such nexuses come into being, transform, and often produce politically ambivalent results. Relatedly, we will also have an eye on how such forms of political knowledge are connected to the claims of professional ‘jurisdictions’ (Abbott, 1988) through which specific social groupings – e.g. statisticians, food planners, social scientists – have sought to establish their role as advisory experts.

Contesting the city

Politics is not only about the modes of operation and reproduction of power constellations; it is also about the social and definitional struggles through which such constellations are established, contested or negotiated. A second dimension of the politics of urban knowledge that is addressed in this book is therefore the co-production of urban knowledge and political conflict. Urban conflicts have long been a central theme of urban history, but they have been mostly described in terms of conflicting interests or competing values rather than as forms of making

and mobilising knowledge (see e.g. Roberts, 2009). The contributions in this book do this in two complementary ways.

On the one hand, they study forms of knowledge as resources that historical actors have mobilised to make claims about, for example, the existence of certain urban problems, how these problems should be addressed, and how they form alliances and organise their strategies of collective political action. Knowledge is mobilised by actors or publics who contested hegemonic discourses and practices of urban governing (e.g. Sandercock, 1998; De Munck, 2014, 2017; Van Damme and De Munck, 2017; De Munck and Romano, 2019). Our volume brings into focus how different kinds of knowledge have gained traction as political resources, and how they have actually been articulated in urban conflicts of different periods, and also how hierarchies between different types of knowledge (between professionals, as well as between professionals and publics) are maintained.

On the other hand, the contributions are sensitive to epistemic conflicts, i.e. conflicts in which knowledge and the authority to define such knowledge are not only a resource of political-claim-making, but are themselves the object of contestation. From a science and technology studies (STS) point of view, a key political dimension of knowledge is its openness to interpretation and negotiation. At the same time, claims about facts and views are intrinsically interrelated to claims about who can be seen as the expert and who is legitimised to speak the truth about the issue (Gieryn, 1999).

As already hinted at above, this is related to the Eurocentric character of modernisation processes and the processes of professionalisation and of building hierarchies between different types of knowledge in which it takes shape. A standard ingredient of modernity is that abstract, standardised, and statistical knowledge as well as knowledge related to geometrical representations are seen as ranking above vernacular types of knowledge related to people's daily experiences and discourses. While this has political implications and has a profound impact on the specific types of governmentalities emerging anywhere in the nineteenth and twentieth centuries, it is also specifically connected to the uneven power relation between Western and non-Western regions.

Imagining the 'good' city

Third, the book sees knowledge-making as both based on and productive of political visions and imaginaries of the good city and related problematisations of urban conditions. Institutionalised modes of governing, as they have been discussed, revolve around such visions, as do the knowledge practices of heterodox urban professions and civic movements.

Although urbanist visions are a long-standing theme in planning history (e.g. Hall, 2014), such literature has tended to treat them mainly as

normative ideas of visionary individuals, movements or professions but not as an active dimension of urban knowledge-making. A related problem is that such work has tended to see urban visions as direct reactions to urban problems, which ignores how imaginaries themselves can actively shape views of problems (Angelo, 2020). In this book we see imagination not only as an outcome, but also as a guiding dimension of the processes of knowledge production, which actively helps shape certain framings and articulations of problems and related strategies of change (see also Munck and Bellavitis, 2017; Pinder, 2005).

This is very much in line with recent work in science and technology studies on what Jasanoff and Kim (2009, 2015) have called ‘socio-technical imaginaries’. These are ‘collectively held, institutionally stabilised and publicly performed visions of desirable futures’ which connect shared understandings of good social life and social order to the pursuit of techno-scientific innovation (Jasanoff, 2015: 4). Such imaginaries can be seen as critical sites of the ‘co-production’ of social and technomaterial orders. Originally coined for comparative science-policy analysis, this perspective has also been extended to works on technological innovation and infrastructure planning in cities (Miller, 2020; Parks, 2020; Sadowski and Bendor, 2019). While such work primarily focuses on imaginaries or ‘visions’ (constructions of attainable futures that are not yet widely shared), the contributions in this volume take a wider perspective on complex urban knowledge processes, which include vision-making as one important element. They treat urban visions and imaginaries as motivating and orientating frameworks for surveys and other forms of urban enquiry, as well as practically and discursively constructed forms of anticipatory and evaluative knowledge.

Even if they do not use the very term ‘imaginary’ or refer to Jasanoff, almost all chapters in this book take into account how certain ways and local trajectories of knowing the city go hand in hand with such new ways of imagining what a ‘good city’ is, and how the city, therefore, should best be governed and designed.

Circulating knowledgeable politics

Although knowledge is embedded in the local political configurations of specific cities, these cities and their knowledge processes are not isolated. This is why urban planning historians have long been interested in the international ‘diffusion’ (Ayatac, 2007; Ward, 1999), ‘transfer’ (Gantner and Hein-Kircher, 2017) or ‘circulation’ of planning ideas and practices (Hård and Misa, 2008a; Harris and Moore, 2013; Hebbert and Mackillop, 2013; Jacobs and Lees, 2013). Such work has focused on which ideas travelled where, along which pathways (conferences, key personalities) and with what consequences. Lissa Robert’s (2009) views on the circulation of knowledge in science, or John Urry’s (2000)

sociological conceptualisation of global mobilities as ‘flows’ and ‘scapes’ traced such processes in other domains. As with the other three themes, such circulation is a process in which the making of knowledge claims is inextricably connected to the articulation of political concerns and agendas. Along with urban knowledge, the problematisations, the governmental techniques and the imaginaries also travel and vice versa. Moreover, such processes are political in a broader sense too, as they tend to be structured by global power relations, e.g. between the ‘West’ and the colonial or postcolonial countries or within different European cities. It is also important to note that such knowledge exchange is not restricted to professional urban expertise, but that urban movements also thrive on translocally connected modes of learning and knowledge exchange (McFarlane, 2011).

In their work on urban technologies, Hård and Misa (2008b) have aptly described this as a dialectic process of ‘circulation’ and active ‘appropriation’ of circulating items within distinct localities. With ‘circulation’ they refer to the exchange of problem definitions, technological knowledge and artefacts that link urban experts across distant places around the world. With ‘appropriation’, they describe the ways in which such circulating matters are incorporated into the planning discourses and material form of particular sites. Appropriation is always a selective and transformative process through which circulating concepts and objects are adapted to local traditions.

While Hård and Misa focus on technology and related skills, such a perspective can also be useful for an understanding of the dynamics of other forms of urban knowledge. Thus in their introduction to a journal section on East European cities, Gantner and Hein-Kircher (2017) describe urbanisation as a transformative knowledge transfer that was guided by specific contextual factors in this geographical region that set it apart from the Western European cities on which much planning was focused (due to its multiethnicity, multiconfessionalism and specific types of nation-building).⁵ Indeed their view of ‘emerging cities’ as dynamic and mutually interacting ‘spaces of knowledge’ comes very close to our processual understanding of urban knowledge.

Urban policy studies and also some work in urban planning studies (Peck and Theodore, 2010; McCann and Ward, 2010; Healey, 2013) have taken a quite similar approach to describe how certain ‘policy models’, such as the ‘Barcelona model’ of urban regeneration (McCann and Ward, 2010), have travelled to other cities and have transformed them as a result. In contrast to conventional understandings of the geographical ‘transfer’ of allegedly stable policy and planning paradigms, they have applied relational perspectives of mobility studies, which see the moving entity as continuously changing in such processes. Peck and Theodore (2010) justify their interest in the mobility of urban policy and policy knowledge with the acceleration of such processes

under contemporary conditions of ‘reflexive governance’. In contrast to the widespread use of the term ‘diffusion’ in historical studies of the international spread of planning ideas, conceptions such as ‘appropriation’ and ‘circulation’ or ‘mobility’ require a more detailed view of the processes-dynamics of knowledge.

Our volume concentrates on various instances of such relational circulation of knowledge and the ways in which circulation transforms the knowledge in question. In contrast to existing studies on policy and knowledge circulation, however, our concern is not so much the process of circulation as such, but rather how it contributed to the formation and transformation of specific configurations of urban knowledge. Furthermore, we are not only interested in general ‘ideas’ but more in how the incorporation of translocal elements transforms other elements of the urban knowledge processes in specific cities.

Overview of the volume

The chapters of this book can be read as elaborations of these four themes, and their various interconnections. They do this by employing quite different approaches and sets of concepts, and they also vary with regard to which of the themes they put most emphasis on. While all chapters share a processual view on urban knowledge and its political articulations, the first set is primarily interested in the identification of broader historical-epistemological patterns, or ‘ways of urban knowing’, by examining how they came into being or by following their longer-term development. A second group of essays, by contrast, traces the local trajectories of urban knowledge processes, and seeks to understand their epistemic and political implications within the context of distinct cities.

Ways of urban knowing

‘Ways of urban knowing’ we understand here as long-term patterns in the production and validation of urban knowledge.⁶ Although the authors in this section employ different terms, they share a similar interest in such patterns, and build their historical narratives around their contrasts or evolutionary dynamics.

In Chapter 2, Jasna Seršić and Bert De Munck describe the period between the Age of Antiquity and the Renaissance as a phase of fundamental change in the ontological frameworks through which people made sense of the city and its politics. In the Greek polis and in medieval European cities, knowledge of city-building and related urban imaginaries were inextricably tied to the practical skills of craftsmen and hands-on workers, who enjoyed considerable epistemic authority. Seršić and De Munck confront this with what they call ‘cartographic reasoning’, an epistemic framework that can already be identified in Vitruvius’

architectural writings, but which eventually gained traction during the Renaissance. Intellectual ideas gradually took precedence over practical ‘making’ as a source of knowledge and authority, and a city was gradually imagined as a geometrical abstraction of points and lines. The authors interpret this as a formative moment for the epistemological logic that has also guided later attempts in urban planning and Western ways of thinking. Basically, this means that the idea and the plan precede, and are authoritatively privileged to, mere making and execution. This politico-epistemic shift is not taken for granted as a natural result of some intrinsic logic of cognitive rationalisation; rather, Seršić and De Munck consider it grounded in the locally and historically specific socio-political conflicts and political philosophies of late medieval and early modern cities. The epistemological transformation is, thus, shown to be deeply entangled with transformations in how the city is shaped and imagined – i.e. with the city as an ontological reality.

As Chapter 3 makes clear, the nineteenth century, and the establishment of urbanism as a discipline, led to a further differentiation of such ways of urban knowing. Wouter van Acker and Peter Uyttenhove take a broad view on European countries in this period, proceeding from the observation that the changes in the governmental and public meaning of urban knowledge were fundamentally entangled with such urban modernisation processes as the industrial revolution and the related social and hygienic issues. They distinguish three different ways of urbanist knowledge-making, each of which configured the city as an epistemological field of urban governing and imagination and each of which was a key ingredient in Western modernity: the discourse of urban reform, the survey and the preservation of heritage. They also show how each of these three registers was tied to specific modes of public display which sought to shape corresponding civic sensibilities: the Parisian *Musée Social*, Brussels’ ‘*Urbaneum*’ and the German ‘*Altstadt*’. Uyttenhove and Van Acker point out that the circulation of knowledge is embedded in specific institutions, which not only develop knowledge but also make knowledge and its related normative imaginaries visible, public and available, and thereby grant legitimacy to new expert-driven forms of urban planning. Relatedly, they show that such institutions and material spaces serve as places of documentation, exchange and diffusion and as material archives in which the city’s historical and transformative identity is not only preserved but is made knowledgeable as well. While this is in itself a precondition for circulation, the institutions are themselves often part of or shot through by international networks.

The emergent ways of knowing addressed in these first two chapters are shown to have shifted the epistemic power to new expert groups, and in particular to have paved the way for the modernist and technocratic forms of planning of the mid-twentieth century. The next two chapters discuss how in the post-World-War period, the rise of new urban

conflicts and alternative imaginaries of the good city have led to far-reaching challenges for these ways of knowing. In Chapter 4, Tim Verlaan and Stefan Couperus describe such shifting modes of urban knowing, or ‘knowledge regimes’ as they call it, in the planning of post-war Dutch cities. The authors show how planners, architects and social scientists managed to gradually dismantle the modernisation paradigm in urban planning and the related ‘scientised’ and elitist ‘knowledge regime’, which in the mid-twentieth century had led to socio-spatial essentialism and to modernity becoming a social imperative of sorts. With the help of activists and civil society organisations they managed to gradually replace the deductive scientific procedures behind the zoning of urban functions and the separation of traffic streams with a more inductive approach, which is sensitive to pre-existing social norms, structures and practices. While this too amounts to a shift in the ontological conception of the city, one in which the city is either an abstracted reality that can be planned from scratch, or a living social organism inscribed in spatial patterns and physical structures, Verlaan and Couperus show that shifts in knowledge regimes are specifically contingent on social norms and discursive claims about urban governance. Building on the Foucauldian idea that power and knowledge are inextricably linked, they actually point at the importance of ‘norm entrepreneurs’, who introduce new norms in policy-making and the production and use of the knowledge involved in it. Expert knowledge related to planning implied social values and norms and specific definitions about welfare and well-being (thus ‘imaginaries’) based on the nuclear family and the functional separation of living, working, recreation, etc. The contestation of these planning practices and the knowledge involved implied a contestation of these norms and values and specifically resulted in a shift from a technocratic and ‘scientised’ to a ‘sociocratic’ planning regime.

Trajectories of urban knowing

The second set of chapters traces how urban knowledge processes unfold over time in distinct local contexts and details how they are co-productively entangled with the formation and governance of major urban-political issues. They also explore how such issues were tied to changing imaginings of the ‘good city’, and how they formed arenas of knowledge-based urban contestations. Moreover, and to a varying degree, they trace how the knowledge and politics that are performed in the studied cities were articulated in wider networks of trans-local circulation and appropriation.

In her case study of a mid-nineteenth century statistical survey in Brussels (Chapter 5), Kaat Louckx examines the role of statisticians such as Adolphe Quetelet in the making of cities as an imaginary and governable reality. While statistics are in themselves an important way of

urban knowing that form part of the survey-orientation that Uyttenhove and Van Acker discussed in section I, Louckx shows how a standardised representation of the Brussels' population took shape in a local trajectory of collective work that was coordinated between Quetelet and a commission of urban and national officials. The selection of themes and methodological choices relied as much on the mutual alignment of the interests of these actors as on the interaction with the interviewees, who sometimes refused to participate in the survey or who gave incomplete answers. At the same time, Louckx details the connections through which the work of the census was articulated with the wider socio-political context, which in nineteenth-century European cities included the working and hygienic conditions of the urban labouring poor, and in the case of a capital like Brussels (the pioneering city in terms of statistics in the mid-nineteenth century) the field of tension between the city and the nation. It is interesting to note in this respect that urban statistics have in the end even played a role, according to Louckx, in the nation-building project of Belgium, referred to by her as 'peasants into Belgians'. Another relevant context is the one in which politics revolved around productivity, which in the case study of Louckx is exemplified by the attention to the 'vitality' of the population, such as the ratio of the active population as charted through the numbers of children and adults.

Like statistics, zoning can be seen as a major component of expert-based ways of urban knowing, as they have evolved since the rise of the Renaissance's 'cartographic reasoning'. In Chapter 7, Julio Paulos and Marko Marskamp also take a local perspective on the trajectory in which such a type of knowledge-making has unfolded. Taking Vancouver as a case study; they detail how, since the early twentieth century, the city became divided into districts that were categorised by use, density and form and how such land-use zoning framed debates and policies of urban development. Their chapter moreover identifies how the instrument of land-use zoning was continuously adapted over time to meet changing socio-political concerns and contexts. Their argument is that this is not to be seen as a technology that either responds simply to the changing context or produces certain effects in the name of certain interests; it also shapes the actors and subjects of planning involved. The plans and devices related to this type of planning ultimately function as 'an interface for plural engagements, negotiation, and validation' around which power relations crystallised and which helped install ways in which to govern the city and steer urban development.

While the circulation of knowledge has been an important dimension of all these cases, Chapters 6 and 8 take a more systematic look at the interaction of urban knowledge cultures in Europe and notably North America on the one hand and those in the 'Global South' on the other. In Chapter 6, Patricia Capanema, Alvares Fernandes and Viviana d'Auria trace the development of urban planning in nineteenth-century Brazil, addressing with

Bela Horizonte a paradigmatic case of Latin American planning history. Fernandes and d'Auria question earlier interpretations of the knowledge practices behind the construction plan for Belo Horizonte, in which the circulation of knowledge was seen as a linear process of knowledge transfer, proceeding from paradigmatic examples in other countries such as Washington, DC. They specifically show that the planning technique of the grid, upon which these plans were based, was transformed and subverted through active processes of appropriation in the local spatial and socio-political context. Their focus on the 'assemblage of practices' reveals an extremely broad range of factors which were involved in the adaptation and appropriation of the abstract gridded plan, even in the case of a new city. The need to accommodate the popular classes, preoccupations about ethnicity, migration and the need for food production eventually resulted in a fundamental blurring and even subversion of the zoning logic inscribed in the plan, including the mingling of urban, suburban and rural functions. This is what emerges when research looks 'beyond the plan' and includes not only a broader range of actors, but a broader range of sources as well (think about laws, cadastral maps, census data, newspaper narratives, etc.). The plan itself in this chapter can also be seen as an 'actant' of sorts, which can foster dialogue and interaction.

In a similar vein, Pouya Sepehr and Erik Aarden (Chapter 8) explore how, in the 1950s and 1960s, urban planners in Iran became increasingly orientated towards US-planning knowledge and related imaginaries of Western urbanism. As they show, Iran was particularly conducive to the appropriation of US-based expert knowledge during the heydays of the modernisation paradigm in the 1950s and 1960s – a trajectory that is largely prefigured by world-wide post-colonial power geometries. Here too, planning knowledge arrived in the form of abstract plans and templates which could be superimposed – at least in theory – from the top down and which disregarded, sometime in overt Orientalist terms, the vernacular history and customs of existing cities. Similar to what Fernandes and d'Auria observed for nineteenth-century Brazil, Sepehr and Aarden point out that the Western tendency to see cities as abstract places (not unlike what Seršić and De Munck dubbed 'cartographic reasoning') was challenged by the views of Iranian planners, for whom the history and complexity of the city mattered in an ontological way. While the former view materialised in master plans that could easily circulate from one place to another, the latter type of knowledge remained tied to a locality. And again, the two different cultural and epistemological repertoires were entangled in processes of negotiation, resulting in processes of adaptation and appropriation. Sepehr and Aarden insightfully point out that streets too, whether they were rendered on a city map or physically present in the urban landscape, functioned as 'boundary objects', which both connected and kept separate the discursive worlds of US and Iranian planners.

Nor, of course, were processes of negotiation and contestation limited to the post-colonial context. In Chapter 9, Sonja Dümpelmann takes us to Berlin, where in the 1980s and 1990s new problematisations and imaginaries of the role of urban greening entailed changes in the knowledge and governance of street trees. Dümpelmann focuses explicitly on how urban knowledge was negotiated at the interface of civic practices, science and urban administration. This was a period in which socio-environmental factors such as drought, road salts, acid air and even dog urine, were identified as causes of the worsening health of street trees. Dümpelmann shows that Berlin tree protesters articulated not only aesthetic and normative preferences, but also knowledge claims about the state of street trees, their mismanagement by the public authorities and about how they could be cared for more effectively. In so doing, they challenged official administrative and expert discourses. They did not pursue a completely alternative epistemology, but rather forged alliances with alternative expert groups. As Dümpelmann argues, the governing techniques to care for and regulate street trees were the outcome of interactions between environmental activists, scientists and policy makers. Moreover, these interactions resulted in – and were in turn mediated by – technological devices such as aerial photographs, maps, index cards and computer lists, the history of which is profoundly entangled with the power of the actors involved, while in turn having an impact on the power distribution as well. When it comes to the democratising effects of this interaction, however, Dümpelmann describes quite ambivalent results. On the one hand, activists had raised awareness about and concern for the environmental problems encountered by trees through lobbying and actions targeting visibility in popular media and the public sphere. On the other hand, scientific research and the development of tracking and charting techniques shifted the power balance in favour of experts. In this process, the trees themselves also transformed, turning from real (individual) trees into ‘bureaucratic’ trees. While the latter were understood as trees which can be ‘engineered’, the real tree was imagined in a more holistic way, and therefore partly beyond human control. Arguably, this can be framed in a ‘sub-altern’ perspective, in which power relations are contingent on which type of knowledge is considered legitimate or not.

In Chapter 10, Jens Lachmund also focuses on the knowledge politics of urban environmentalism and on the role of civic activism in the shaping of urban knowledge. He traces how since the 1970s London has become a centre of urban agriculture in Europe, first with some experiments in community projects such as collective gardens and city farms, and later through the formation of a distinct expert field of ‘food planning’, related policies and imaginaries of the ‘good city’. New discursive and political orderings of the city as a ‘food system’, and the problematising claims that come with this concept, were not pre-given

entities around which publics and policy makers in London constructed their concerns. It required knowledge-making by food planning experts and policy makers as well as the civil society members who interacted with them to make these objects visible and accountable and to prepare them as targets of public concerns and politico-administrative intervention. This knowledge-making included the crafting of definitions and classifications, the collecting and combining of data of different sorts and the organisation of these findings through narratives that resonated with wider public concerns. Rather than simply describing a given state of the city, these knowledge-practices performed a new epistemic and ontological order of the city, and thereby opened up a new space for political deliberation and policy-making. As Lachmund shows, the actors who were involved in this new planning field not only pursued common knowledge claims about the need and viability of urban agricultures, but forged fragile alliances across different communities and epistemic practices. While gardening activists relied on their local practical experience in projects, food-planners relied on survey techniques, which extended established practices of the urban survey. Those who actually managed the campaigns in which food-growing was promoted in London combined techniques of public citizen mobilisation with epistemic practices of counting and visibilisation, which targeted the citizen both as an object and actor of urban knowledge-making.

Jointly, the chapters by Lachmund and Dümpelmann make clear that different ways of knowing the city do not have to be incompatible; rather they describe how the knowledge involved in political negotiations between bureaucrats, planning experts and citizens transformed – resulting in changing power balances and alliances between experts and lay-activists.

Tying in with this, the collection ends with Chapter 11 by Andrew Karvonen, which both historicises and localises the arguably most far-reaching turn in the recent urban history of knowledge: digitalisation and the related imaginary of the ‘smart city’. From the early 2000s, Manchester became entangled in regional science policies and a regional economic development agenda of the central state that shifted the field of gravity to the regional level. His case study traces the origins and politics of Manchester’s twenty-first-century smart city as an urban knowledge process that proceeds simultaneously at the levels of storytelling and boosterism, new mechanisms of urban governance and material urban form. Karvonen reveals the role of entrepreneurial elites, policy makers and knowledge experts and shows how they are entangled in what he calls ‘a triple helix’ dynamic. Jointly, they tend to dominate the urban agenda to the detriment of grassroots initiatives, local communities and civil society organisations. This is all the more invasive as the dominant actors not only join in public-private partnerships, but also cultivate a discourse of innovation and modernity in which the industrial and

technological history of the city is itself implicated as a justification and source of inspiration and in which knowledge formation is seen as a driver of history. This brings us full circle to the debate on imaginaries about modernity. The chapter shows that knowledge and the way it is involved in urban developments is contingent on narratives about modernity, improvement and the pioneering role of specific Western cities.

Conclusion

By following the politics of urban knowledge throughout the rise of Western modernity, our book has revealed the diversity, hybridity and local contingency of the work through which planning experts, policy makers and critical activists have made sense of their city, and through which they have constructed claims about how to make them better. Its contributions do this through the format of detailed case studies, each of which reveal quite distinct configurations of power, knowledge and city-making.

The chapters have revealed that the various turning points described by urban and planning historians, such as the rise of the industrial city, the colonial expansion of urbanism or the increasing concern for sustainability, have always been moments of distinct epistemic productivity. They were episodes in which established concepts and categories used to capture the city and its problems were challenged by new frameworks of knowing. Rather than just reflecting objective conditions, such as poverty, bad housing conditions, or environmental problems, they were key in shaping these realities in socially meaningful forms. They were also moments of shifting epistemic authority that allowed new actor groups to come to the fore. This invites us to see transformative moments in urban history not simply as the self-evident results of modernisation, or of the articulation of new interests. What contemporary actors considered as their city, what they imagined as modern or sustainable, and how they articulated their visions and interests about the 'good city' was driven or at least mediated by knowledge processes. This does not mean that knowledge-making was an independent factor of social change; it always worked in conjunction with other socio-political forces and circumstances.

The book also reveals a complex interplay between continuity and change in urban knowledge that runs counter to any linear understanding of modernisation or of the rise of one single dominant logic of governmentality. On the one hand, we see that changes build on each other, reproducing epistemic schemes and practices in situations which can be quite different in other respects. So, there is clearly a continuity in the logic of abstraction and standardisation that connects Renaissance cartographic reasoning with the nineteenth-century sanitation programmes, the twentieth-century modernist planning practices and the twenty-first century digital managerialism of smart

city projects. At the same time, the case studies reveal that abstraction is tied to quite different practices and agendas in each of these projects, and that this gives them very different meanings and political values, depending on time, place and the issue addressed. Moreover, we have seen that tangible moments of urban knowledge-articulation typically mix or confront diverse ways of knowing the city. What emerged in this volume is a plurality of urban knowledge both in the professional world and in bottom-up, vernacular and subversive forms of knowledge-making arising outside these professional worlds. With respect to the latter, we have emphasised their innovative character, but it might be equally valuable to draw attention to long-term continuities, for example between the identification of knowledge with ‘making’ in Antiquity and medieval cities, on the one hand, and the practical learning of activists in contemporary community gardens, on the other. Rather than viewing urban history as the increasing dominance of just one way of knowing the city, our collection of essays offers a view of urban knowledge history in terms of overlapping and coexisting modes of knowing the city.

Notes

- 1 On the performativity of knowledge see Mackenzie (2008), who has shown how theoretical models of finance have played an active role in the construction of the financial markets that they describe (by making them more similar to how they are described by these models).
- 2 There also exists interesting work by a network of historians of science and of architecture on the ‘epistemic history of architecture’ which has studied the formation and circulation of skills in building practices from antiquity to renaissance (e.g. Bührig et al.; see also the related series of publications in German <https://www.mpiwg-berlin.mpg.de/resources/publications/books/0206>). Urban knowledge as we discuss it here, can include architectural knowledge, but refers more broadly to the performative role of knowledge in the shaping and governing of entire cities.
- 3 Madden’s contribution is part of a longer debate in the journal *City*, which was spurred on by Schafran’s (2014) call for a more active position of urban study scholars, and which also included contributions by Brenner and Schmid (2015), Walker (2015) and Meagher (2015) on the epistemological underpinnings of urban studies. For a more theoretical-epistemological treatment of urban knowledge, see also Dell (2016).
- 4 Jasanoff (2004) introduces ‘co-production’ as an ‘idiom’ of empirically oriented theorizing which focuses on mechanisms and processes that systematically connect scientific and technological knowledge-making with the ordering of politics and social life. This includes both a ‘constitutive’ understandings of nature and society in the tradition of actor-network theory, and her own ‘interactive’ view of knowledge-making and socio-political ordering.
- 5 See also the case studies in the special section on this theme that is introduced by their essay: *Journal of Urban History*: 2017, 43(4).
- 6 In contrast to Pickstone (2000), who has identified five global ‘ways of knowing’ (‘world reading’, ‘natural history’, ‘analysis’, ‘experimentalism’ and

‘technoscience’) in the development of modern science, we do not aim to provide a comprehensive map of ‘ways of urban knowing’. We use the term more pragmatically, to characterise the focus of the contributions in this section of the book.

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Introduction: Urban knowledge and the politics of governing cities

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Smart cities, knowledge generation and the enduring pursuit of urban innovation

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